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Portfolio Selection of KOMPAS-100 Stocks Index Using B-Spline Based Clustering

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Abstract

Investment risk in stocks is one of the things that need to be considered by investors. Therefore investors need to develop strategies to manage portfolios. One way to manage risk in stock investments is to diversify the portfolio by selecting stocks. In this paper, stocks in a portfolio are chosen based on the similarity of the price movement data through the clustering using K-means. Since stocks in the same cluster have a higher similarity compared to shares in different clusters, the portfolio consists of stocks selected in each different cluster. Stock price movements are high dimensional data, requiring computation costs during clustering, so the dimension reduction is applied by conducting an interpolation using B-Spline. Based on the weekly dataset in 10 years (01/01/2009 - 12/31/2018), the Mean-Variance and the Equal-Weight portfolio consists of the selected stocks using this approach offer less volatility, higher Sharpe Index, and better cumulative performance.

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Keywords: portfolio, b-spline, clustering, KOMPAS 100, mean-variance, equal-weight;

1. Introduction

Research on portfolios with various methods has long been carried out, one of which is how to get a minimum risk portfolio. To obtain the minimum risk, stocks in the portfolio need to be well-diversified, one easy way to look at it is if the pattern of price movements varies.

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There are many methods to form a portfolio. The most commonly used are the Mean-Variance (MV) portfolio (the pioneer), Equal-Weight (EW) portfolio (the simplest), and Equally-Weighted Risk Contribution (ERC) portfolio. [1] However, building a portfolio that involves a large number of stocks is not efficient. Therefore, the selection of stocks to be included in the portfolio is very necessary. There are various methods for stock selection, but almost all of the selection conducted based on stock returns, such as Pattarin et al. (2004) [2], Nanda et al. (2010) [3], De Luca and Zuccolotto (2015) [4], and D'urso et al. (2016) [5].

Another approach is to select a stock portfolio based on the similarity of the price movement curve, as conducted by Silva and Marques (2010) [6]. The simple way to see the pattern of stock price movement is by identifying time intervals in which the stock price is up and down from historical data. Then the similarity of the pattern of stock price movement is identified by conducting clustering. However, dealing clustering with the original stock price movements will lead us to high dimensional data. To overcome these issues, Lorio et al. [7] proposed a time series clustering analysis on P-spline coefficients that reduces dimensionality and ensures computational time-saving.

We propose dimension reduction data by sampling stock price data, and then we do an interpolation curve using B-spline. So far, few studies have compared the performance of P-Spline and B-Spline. For the same number of knots and orders, the P-spline output variable is slightly more than the B-spline, because the P-spline contains a penalty parameter. Based on our experiments, the B-spline estimation for a sufficient number of knots gives better results than P-spline. After conducting K-Means clustering, we construct a portfolio consists of stocks chosen from a different cluster to keep diversity in the portfolio. The weight of the portfolio determined by Mean-Variance and Equal-Weight methods.

2. Related Work

The pioneer of modern portfolio theory is Markowitz that proposed the Mean-variance (MV) portfolio [8]. The weights of the MV portfolio determined by considering expected return and risk that usually measured by the variance of return. [9] Another alternative to determine the portfolio weight is by setting all weights equal, namely all weight equal to$1/N$, where N is the number of stocks as proposed by Benartzi and Thaler [10]. This kind of portfolio is known as an equal weight (EW) portfolio.

Diversification in the portfolio is a strategy to manage investment risk. In constructing a portfolio, an investor needs to select stocks and determine the weight (proportion) of stocks in a portfolio. Stock selection for portfolios commonly based on their return. Pattarini et al. (2004) [2] proposed a classification scheme to identify mutual funds styles based on past returns. The scheme consists of dimensionality reduction, clustering, and style identification. Nanda et al. (2010) [3] applied some clustering methods on stock return. De Luca and Zuccolotto (2015) [4] applied clustering on stock data according to their possible extreme return. D'urso et al. (2016) [5] proposed a GARCH model to estimate stock volatility and apply a fuzzy clustering.

Silva and Marques (2010) [6] proposed a different base for the portfolio selection who applied Self Organizing Map (SOM) clustering based on the similarity of prices. Lorio et al. (2018) [11] proposed a clustering approach similar to Silva and Marques (2010) [6]. Lorio et al. applied a P-spline to reduce the dimension of stock data of the Italian FTSE stock index and the S&P 500. The selected stocks collected in a portfolio of Mean-Variance (MV), Equal-Weight (EW), and Weight Equal Risk Contribution (ERC).

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3. Proposed Model

The system proposed in this paper is shown in figure 1:

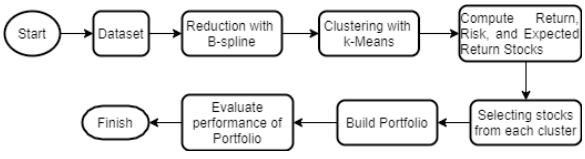

Fig. 1. The System Process

3.1 Dataset

The data used in this research is the weekly close price data of the KOMPAS 100 stock indexes from 01/01/2009 until 12/31/2018, the data obtained from yahoo.finance.com. The dataset consists of 66 stocks of 100 stocks belonging to the KOMPAS 100 index. Each stock data has 522 data rows, of which 522 are obtained based on the number of weeks of a month that produces 4 to 5 data rows and 120 months. The list of stocks given in Table 1.

Table 1. Index KOMPAS 100

No	Initial	Assets Description
1.	AALI	ASTRO AGRO LESTARI
2.	ADHI	ADHI KARYA
3.	ASII	ASTRA MOTOR
4.	BBCA	BANK CENTRAL ASIA
5.	BBNI	BANK NEGARA INDONESIA
…	…	…
100.	WTON	WIJAYA KARYA BETON TBK.

3.2 B-spline for Dimensionality Reduction

B-spline has segmented properties, and the formula of degree p is defined recursively as [12]

$$B_{i,p}(t) = \frac{t-t_i}{t_{i+p}-t_i} B_{i,p-1}(t) + \frac{t_{i+p+1}-t}{t_{i+p+1}-t_{i+1}} B_{i+1,p-1}(t), \quad (1)$$

where the set $\{B_{i,p}(t), i = 1,2, \dots, n\}$ is a B-Spline basis.

The original dataset has 522 dimensions and needs to reduce. Firstly, we sample the dataset with 15, 25, 35, 45, 55, and 65 records. Then we apply B-Spline to the sample with corresponding knots 15, 25, 35, 45, 55, and 65, and RMSE (Root Mean Squared Error) is applied to evaluate the B-spline coefficient in fitting the actual data 10. The chosen knots for this study based on the Equally-spaced rule proposed in Ruppet [13] research and Lorio at el. (2018) [11].

3.3 K-Means Clustering

The next step is clustering that aims to group the B-Spline coefficient of each stock based on their similarity using the K-means method. First, we applied K = {2, 3, 4, 5, 6, 7, 8, 9, 10}, intending to determine the effect of the number of members on the portfolio against performance stock portfolio, and applying K-Means to a group of P objects will the objects into K cluster. Each cluster has an initial centroid. These initial centroid determined randomly. The K-means algorithm [14, 15] calculates the distance between the centroid and the data in each cluster. This process runs iteratively to update the centroid until it does not change again. The distance function defined as

$$D\,(x, y) = \sqrt{\sum_{i=1}^{p} (x_i - y_i)^2}, \quad (2)$$

where x is the B-spline coefficients, y is the centroid point, and p is the number of data. The centroid y_i is updated by

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$$y_i = \frac{1}{p}\sum_{q=1}^{p_i} x_q, \tag{3}$$

where p_i is the number of data in the cluster–i.

3.4 Building a Portfolio

The most common portfolio is the Mean-Variance (MV) portfolio where the portfolio weights $\boldsymbol{w}$ are the solution of the following optimization problem

$$\min_{\boldsymbol{w}} \boldsymbol{w}^T \boldsymbol{C}\boldsymbol{w}, \tag{4}$$

s.t

$$\boldsymbol{1}^T\boldsymbol{w} = 1 \text{ and } \boldsymbol{w}^T\boldsymbol{\mu} = R^*, \tag{5}$$

where $\boldsymbol{C}$ is the covariance matrix and $\boldsymbol{\mu}$ is the expected returns. Term $\boldsymbol{w}^T\boldsymbol{C}\boldsymbol{w}$ in equation (4) represents a portfolio risk to minimize and R^* in (5) is expected portfolio return required by investor. [8]

Another alternative is the simplest portfolio known as Equal Weight (EW) portfolio [10]. If there are N selected stocks in the portfolio, then the weight of the portfolio is simply

$$\boldsymbol{w_i} = \frac{1}{N}, i = 1,2, \dots, N. \tag{6}$$

3.5 Evaluation of Portfolio

We use back-testing to evaluate the portfolio performance by comparing indicators Sharpe index and volatility [11]. The data test is KOMPAS-100 stock index from 2nd January 2019 to 30th December 2019.

4. Result

4.1 Comparisons Performance of Smoothing Method

To test the model's performance in this research, we compared the smoothing method with the P-spline approach proposed by Lorio et al. 2018 [11]. The Root Mean Squared Error (RMSE) is applied to evaluate the P-spline coefficient in fitting the actual data.

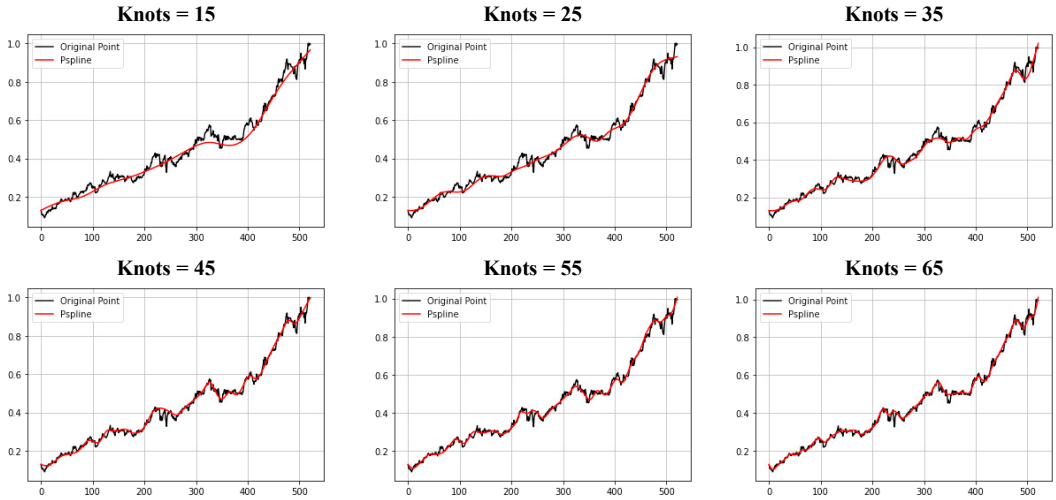

Fig. 2 Smoothing and interpolation with P-spline (solid red line) for a different number of knots. The solid black line is closing price of assets KOMPAS 100. For each subplot, the horizontal axis represents the time and the vertical axis, the price level.

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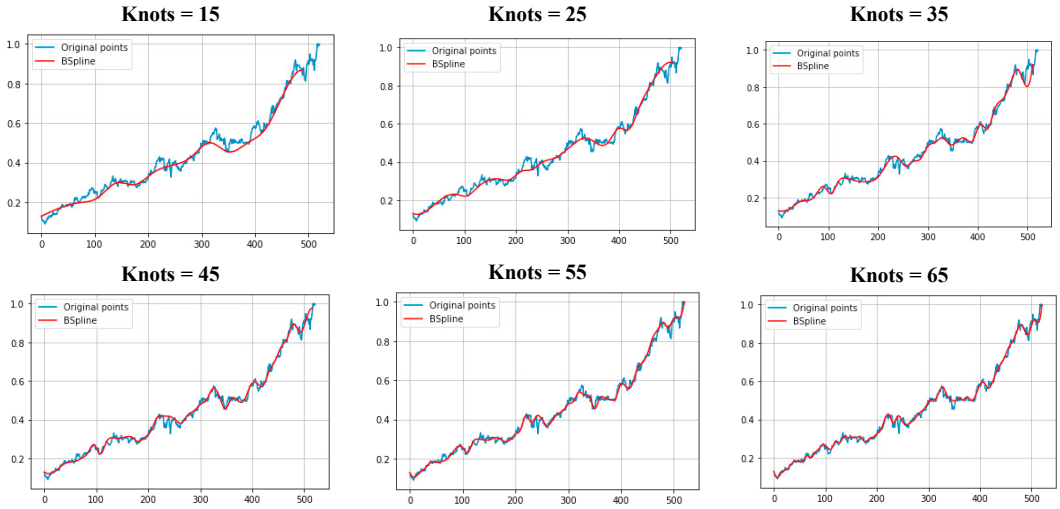

Fig. 3. Smoothing and interpolation with B-spline (solid red line) for different number of knots. The solid blue line is closing price of assets KOMPAS 100. For each subplot, the horizontal axis represents the time and the vertical axis, the price level.

Table 2. Average Table RMSE number of knots applied to smoothing with B-spline

Knots	15	25	35	45	55	65
RMSE	0.1005	0.0754	0.0608	0.0463	0.0367	0.0326

Table 3. Average Table RMSE number of knots applied to smoothing with P-spline

Knots	15	25	35	45	55	65
RMSE	0.0771	0.0589	0.0474	0.0422	0.0388	0.0350

Figure 2 and Figure 3 shown an illustration of the estimated P-spline and B-spline with 15, 25, 35, 45, 55, and 65 knots. Table 2 and Table 3 shown the average RMSE of the P-spline and B-spline of stocks in the KOMPASS 100 index. The performance of both smoothing methods in the same case has evaluated by considering the RMSE value.

The RMSE of the P-spline is less than the RMSE of the B-spline (Table 2) for knot 15 up to 55 knots. Conversely, for larger knots, namely 55 and 65, the B-spline yields smaller RMSE. The difference between smoothing approach with B-spline and P-spline is in the involved of penalties in the process of smoothing with P-spline, which aims to avoid overfitting.

4.2 B-spline Estimation

Illustration of B-Spline estimation using 15, 25, 35, 45, 55, and 65 knots described in Figure 3. Table 2 shows the average RMSE of the B-spline of stocks in the KOMPAS-100 index. The RMSE decreases as the number of knots increases. However, the difference between the RMSE for 55 knots and 65 knots is not significant. Therefore, we choose 55 knots as the standard number of knots in this experiment.

4.3 K-Means Clustering

Illustration of clustering results of 55 knots B-spline described in Figure 4. The proposed portfolio here consists of stocks representing each cluster, chosen from the nondominated set based on their expected return and the variance of return.

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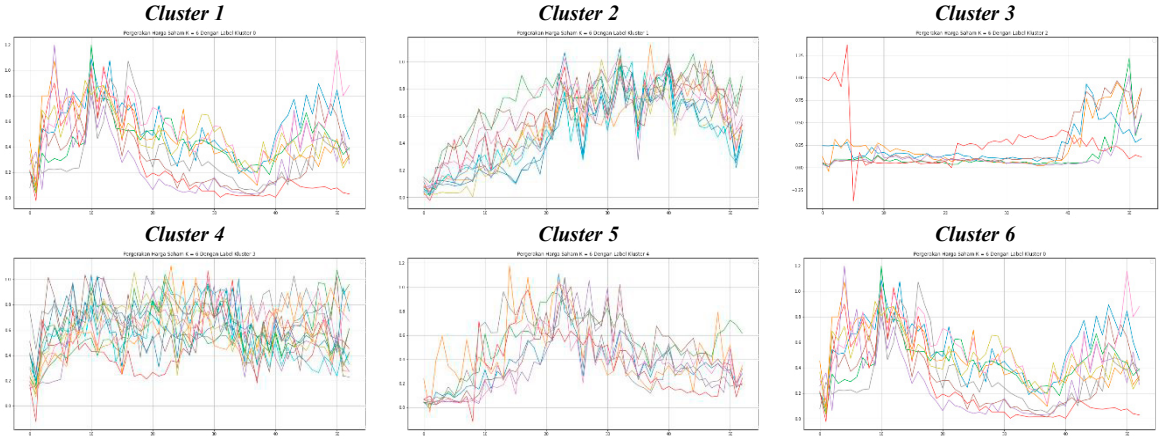

Fig. 4. Time series of the closing price of assets of KOMPAS 100 by K-means. For each subplot, the horizontal axis represents the time and the vertical axis, the price level.

Table 4. The Weights of MV and EW Portfolio with ten stocks.

Assets	MV	EW
UNTR	0.0318	0.1
SMGR	0.2247	0.1
UNVR	0.3660	0.1
TPIA	0.0355	0.1
TINS	0.0677	0.1
SSIA	0.1453	0.1
LPPF	-0.0411	0.1
GJTL	-0.0034	0.1
PTRO	-0.0015	0.1
WIKA	0.1751	0.1

4.4 Building a Portfolio

The weights of the portfolio determined by Mean-Variance (MV) and Equal Weight (EW) method. The portfolio consists of two stocks, three stocks up to ten stocks chosen as a representation of each cluster. The illustration of portfolio weight consists of ten stocks is shown in Table 4. The volatility of the return of the corresponding portfolio given in Table 5. The volatility of the MV portfolio decreases as the number of stocks increases and less than the volatility of KOMPAS-100 (0.000745). On the other hand, the volatility of the EW portfolio larger.

Table 5. Volatility of the Portfolio Returns (01/01/2009 – 31/12/2018)

Portfolio	1	2	3	4	5	6	7	8	9
MV	0.00099	0.00079	0.00075	0.00071	0.00064	0.00063	0.00068	0.0006	0.00061
EW	0.00124	0.00119	0.00143	0.00089	0.00078	0.00092	0.00098	0.0009	0.00107

The expected return of the MV and EW portfolios is positive, as depicted in Table 6. The expected return of the KOMPAS100 is -0.002144.

Table 6. Expected Returns of The Portfolio (01/01/2009 – 31/12/2018)

Portfolio	1	2	3	4	5	6	7	8	9
MV	0.00409	0.00371	0.00407	0.00422	0.00400	0.00400	0.00438	0.00448	0.00476
EW	0.00353	0.00439	0.00349	0.00441	0.00432	0.00425	0.00408	0.00494	0.00513

4.5 Evaluation of Portfolio

The indicators used to evaluate the portfolio's performance are the Expected Return, Volatility, and Sharpe Index. [11] Table 7 shows the expected returns of the corresponding portfolio. Comparing with the expected return of the

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KOMPAS 100 (0.0009255), we could see from the table that almost the portfolio of MV yield higher expected return.

Table 7. Expected Returns of the MV and EW Portfolio (01/01/2019 – 31/12/2019).

Portfolio	1	2	3	4	5	6	7	8	9
MV	0.003369	0.001196	0.000971	0.002478	0.000919	0.001060	0.002434	0.001088	0.001938
EW	0.001282	-0.001294	-0.003243	0.000241	-0.001004	-0.001596	0.000826	-0.001436	-0.001672

Table 8 shows the volatility of the MV and EW portfolio. Comparing with the volatility of KOMPAS 100 (0.000388), the volatility of some MV portfolio is less than KOMPAS 100, and EW portfolios are more volatile than KOMPAS 100.

Table 8. Volatility of the MV and EW Portfolio (01/01/2019 – 31/12/2019).

Portfolio	1	2	3	4	5	6	7	8	9
MV	0.000324	0.000533	0.000379	0.000359	0.000491	0.000490	0.000480	0.000382	0.000380
EW	0.000421	0.001555	0.000993	0.000726	0.000779	0.000913	0.000916	0.000672	0.000905

Table 9 shows the Sharpe index of the MV and EW portfolio. Almost all of the Sharpe index of the MV portfolio larger than KOMPAS 100 (2.3888188).

Table 9. Sharpe Index of the MV and EW Portfolio (01/01/2019 – 31/12/2019).

Portfolio	1	2	3	4	5	6	7	8	9
MV	10.4	2.24	2.56	6.91	1.87	2.16	5.07	2.85	5.1
EW	3.05	-0.83	-3.27	0.33	-1.29	-1.75	0.9	-2.14	-1.85

Fig. 5 Cumulative performance of the portfolio with MV, EW, and KOMPAS 100 (01/01/2019 - 31/12/2019). For each subplot, the horizontal axis represents the time, and the vertical axis represents the value of cumulative performance.

Figure 5 displays the movement of investor wealth changes. We could see from the figure that the MV portfolio grows more stable than the EW portfolio and KOMPAS 100. At the end of the period, MV's cumulative performance value is 1.0844, higher than the EW portfolio, and KOMPAS 100. The KOMPAS 100 is between the two portfolios, and the cumulative performance value is 1.0554.

5. Conclusion

The clustering of stocks based on the similarity of their price movements curves is a promising approach for portfolio selection, to keep diversity in a portfolio. Application of B-Spline to reduce high dimensional data during the clustering process runs well. Based on the experiment using knots 15, 25, 35, 45, 55, and 65, the P-spline gives better RMSE than the B-spline for smaller knots. Conversely, for larger knots, start from knot 55, the B-spline is better.

Optimization of the portfolio weights of the selected stocks is a crucial process. Evaluation using back-testing

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data, the MV portfolios are generally less volatile, higher Sharpe index, and better cumulative performance than KOMPAS 100. A different result is for the EW portfolio, where the weights are not optimized. Generally, their performance is worst than KOMPAS 100 in volatility and Sharpe index.

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